Problem Set 2

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Due: October 22, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 22, 2026. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```r
# Enter observed frequencies by row
observed <- matrix(
  c(14, 6,7,
    7,7,1),
  nrow = 2,
  byrow = TRUE,
  dimnames = list(
    Class = c("Upper","Lower"),
    Outcome = c("Not_stopped",
                "Bribe",
                "Warning")
  )
)

# Calculate marginal totals and sample size
row_total <- rowSums(observed)
col_total <- colSums(observed)
n <- sum(observed)

# Row total * column total /sample size
expected <- outer(row_total, col_total)/n


# Calculate each cell's contribution and sum them
contribution <- (observed - expected)^2/expected
chi_square <- sum(contribution)
```

Table 1: Result of Chi-Square statistic

3.791

H0 : Social class and police treatment are independent

H1: Social class and police treatment are not independent

As shown by the R result, the chi-square statistic is 3.79.

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

```r
# Calculate  df
df = (nrow(observed)-1)*(ncol(observed)-1)

# Calculate p value
p_value <- pchisq(chi_square, df = df, lower.tail = FALSE)
```

Table 2: Result of P Value

0.150

As the p value (0.15) is greater than $\alpha(0.1)$, we fail to reject H0. This means that there is insufficient evidence at the 10% significance level to conclude that there is a relationship between social class and police treatment.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.322	-1.642	1.523
Lower class	-0.322	1.642	-1.523

```r
# Calculate proportions
row_prop <- row_total/n
col_prop <- col_total/n

# Calculate standardised residuals
sr <- (observed - expected)/sqrt(expected*outer(1-row_prop,1-col_prop))
sr <- round(sr,3)
```

(d) How might the standardized residuals help you interpret the results?

Standardised residuals identify which cells depart from the frequencies expected. A positive residual indicates more observations than expected, whereas a negative residual indicates fewer observations than expected. For lower class drivers, bribe requests occurred more often than expected (z = 1.64), while not being stopped (-0.32) and being stopped and given a warning (-1.52) occurred less often than expected. For upper class drivers, the opposite pattern was observed – Bribe request occurred less often than expected (-1.64), not being stopped(0.32) and being stopped and given a warning(1.52) occurred more often than expected.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

[3]Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

Null hypothesis(H0) : The population mean number of new or repaired water facilities is the same in villages with and without the reservation policy.

Alternative hypothesis(H1): The population mean number of new or repaired water facilities differs between the two groups.

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
# Load the data
women <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/
    master/PREDICTION/women.csv")

# Establish the regression model
model <- lm(water~reserved, data = women)

summary(model)
```

water facilities $= \beta_0 + \beta_1 * reserved + \varepsilon$

$H0 : \beta_1 = 0$

$H1 : \beta_1 \neq 0$

Reserved equals to 1 if the village' GP was reserved for woman leaders and 0 if the GP was not reserved for women leaders. The estimated equation is as below: $\widehat{water facilities} = 14.74 + 9.25 * reserved$

(c) Interpret the coefficient estimate for reservation policy.

The estimated reservation coefficient is 9.25. This means that compared with villages whose GP was not reserved for woman leaders, villages whose GP was reserved had, on average, approximately 9.25 more new or repaired water facilities. The p value is 0.0197, which is less than 0.05, therefore we can reject H0.